Security Risk Assessment Report

This is a Security Risk Assessment Report for a <u>fictional</u> scenario given as an activity for the *Connect and Protect: Networks and Network Security Course* in Coursera

Scenario:
You are a security analyst working for a social media organization. The organization recently experienced a major data breach, which compromised the safety of their customers' personal information, such as names and addresses. Your organization wants to implement strong network hardening practices that can be performed consistently to prevent attacks and breaches in the future.

After inspecting the organization's network, you discover four major vulnerabilities. The four vulnerabilities are as follows:

1. The organization's employees' share passwords.
2. The admin password for the database is set to the default.
3. The firewalls do not have rules in place to filter traffic coming in and out of the network.
4. Multifactor authentication (MFA) is not used.

If no action is taken to address these vulnerabilities, the organization is at risk of experiencing another data breach or other attacks in the future. In this activity, you will write a security risk assessment to analyze the incident and explain what methods can be used to further secure the network.

Part 1: Select up to three hardening tools and methods to implement
Password Policies Multifactor Authentication (MFA) Firewall Maintenance, Disabling Unused Ports, Port Filtering Additional tools: Encryption using the Latest Standards Server and Data Storage Backups Usage of SIEM tools and IPS Patch updates

Part 2: Explain your recommendations
Password Policies: A strong password policy ensures that every employee password meets security standards. It should also forbid password sharing and require the default database admin password to be changed. This addresses vulnerabilities 1 and 2. **Multifactor Authentication (MFA):** MFA adds extra layers of defense. If an employee's password is stolen, an attacker still cannot access the account without the additional verification step. This addresses vulnerability 4. **Firewall Maintenance, Disabling Unused Ports, and Port Filtering:** Firewall rules should be set up and reviewed regularly to block suspicious traffic from outside the network. Disabling unused ports and filtering ports also reduce the network's attack surface, which means fewer entry points for attackers. This addresses vulnerability 3. **The following are additional tools and methods that I recommend to further strengthen the network's security, beyond the four major vulnerabilities mentioned:** **SIEM Tools and IPS:** These provide another layer of protection in case an attack gets past the firewall. They detect suspicious activity and help stop it early. **Encryption Using the Latest Standards:** Using current encryption standards, such as a VPN, protects data as it travels across the network and prevents packet sniffing. **Server and Data Storage Backups:** Regular backups allow the organization to recover its data after an attack. **Patch Updates:** Keeping software and operating systems up to date fixes known security weaknesses before attackers can use them.